

Chapter 13---Properties of Solutions

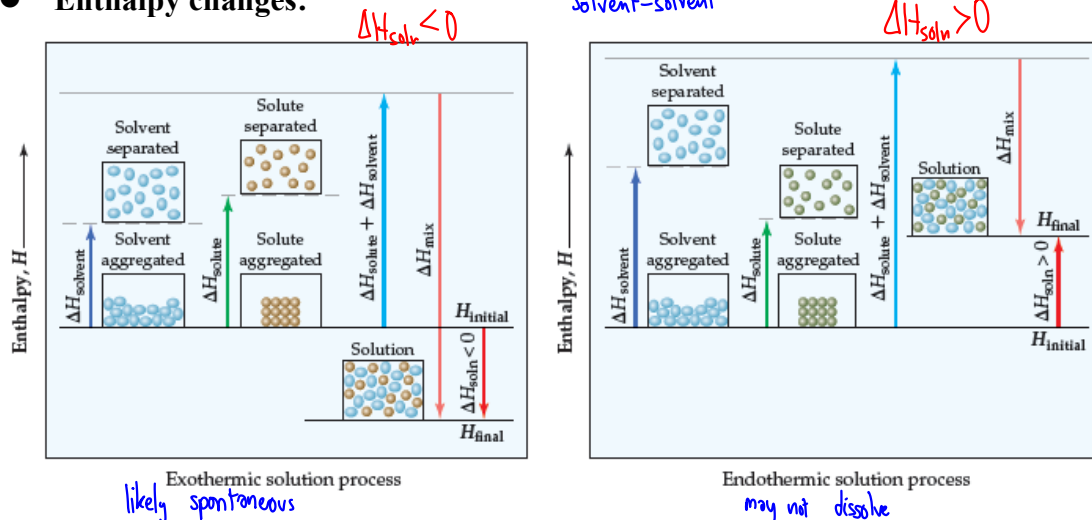
Review

Definitions:

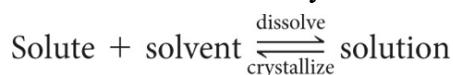
- The ability of substances to form solutions depends on:

- 1) natural tendency toward mixing: similar to gases mixing (spontaneous)
 - 2) intermolecular forces (dispersion force, dipole-dipole force, hydrogen bond, ion-dipole force);
 - ✓ Solute-solute interactions between solute particles must be overcome to disperse the solute particles through the solvent. ΔH_{solute}
 - ✓ Solvent-solvent interactions between solvent particles must be overcome to make room for the solute particles in the solvent. $\Delta H_{\text{solvent}}$
 - ✓ Solvent-solute interactions between the solvent and solute particles occur as the particles mix. ΔH_{mix}
- $\text{solvent-solute} \geq \text{solute-solute} + \text{solvent-solvent}$

- Enthalpy changes:



- **Hydrate:** compounds that have a defined number of water molecules in the crystal lattice.
- **Crystallization:** the process where a solid forms where the atoms or molecules are highly organized in a structure known as a crystal.



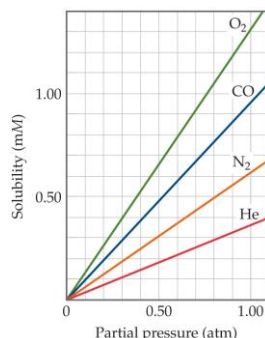
- **Saturated:** a solution that is in equilibrium with undissolved solute is saturated.
- **Unsaturated:** if we dissolve less solute than the amount needed to form a saturated solution, the solution is unsaturated.
- **Supersaturated:** under suitable conditions, it's possible to form solution that contain a greater amount of solute than needed to form a saturated solution, it's supersaturated. *unstable*
- **Solubility:** the amount of solute needed to form a saturated solution in a given quantity of solvent.

- **Factors affecting solubility**

1) **Solute-Solvent Interactions:** “like dissolves like.” Substances with similar ^{stronger} intermolecular attractive forces tend to be soluble in one another.

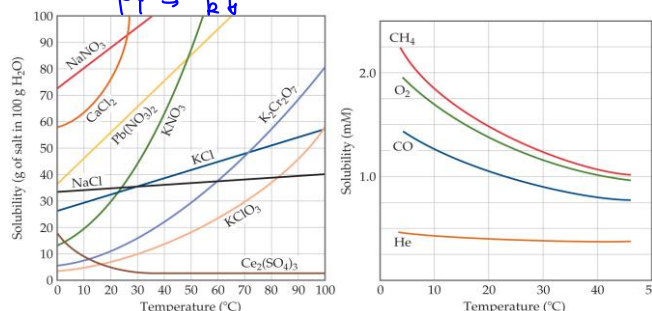
2) **Pressure Effects**----- **Henry’s Law:** $S_g = kP_g$

- the solubility of the gas in a liquid solvent increases in a direct proportion to the partial pressure of the gas above the solution;
- P_g is the equilibrium partial pressure of the gas when it reaches equilibrium;
- The molecular state of the solute should be the same in gas phase and solution;
- The lower the solution concentration is, the more suitable the law is;



3) **Temperature Effects**

- Usually, dissolution of inorganic salt in liquid is an endothermic process;
- Usually, dissolution of gas in liquid is an exothermic process;



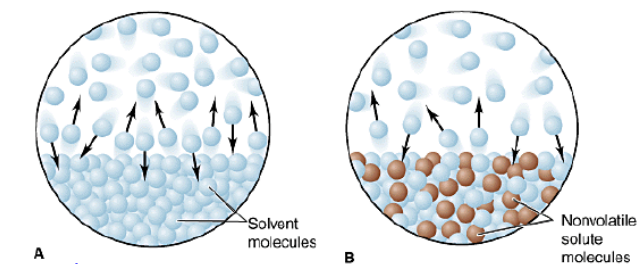
- **Units of Concentration:**

- Mass percentage: mass % of component = $\frac{\text{mass of component in solution}}{\text{total mass of solution}} \times 100$
- Parts per million (ppm): ppm of component = $\frac{\text{mass of component in solution}}{\text{total mass of solution}} \times 10^6$
- Parts per billion (ppb): ppb of component = $\frac{\text{mass of component in solution}}{\text{total mass of solution}} \times 10^9$
- Mole fraction (X): mole fraction of component = $\frac{\text{moles of component}}{\text{total moles of all components}}$
- Molarity (M): molarity = $\frac{\text{moles of solute}}{\text{liters of solution}}$ (do vary with temperature)
- Molality (m): molality = $\frac{\text{moles of solute}}{\text{kilograms of solvent}}$ (do not vary with temperature)

- **Colligative properties:** physical properties of solutions that depend on the quantity

(concentration) but not on the kind or identity of the solute particles.

- 1) **Vapor Pressure Lowering:** the vapor pressure of the solution is lower than that of the pure solvent.

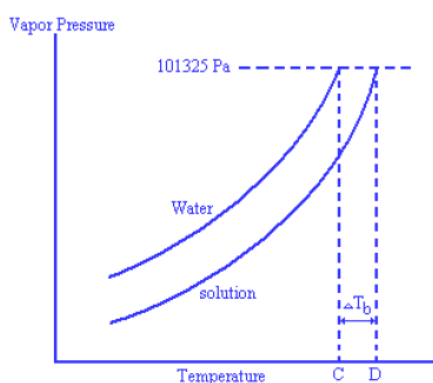


$P_{\text{solution}} = X_{\text{solvent}} \cdot P^{\circ}_{\text{solvent}}$ (Raoult's Law)

$\Delta P_{\text{solution}} = X_{\text{solute}} \cdot P^{\circ}_{\text{solvent}}$

- The vapor pressure of a volatile solvent above a solution containing a nonvolatile solute is proportional to the solvent's concentration in the solution.
- The law is only suitable for the ideal solutions (the size and force of the solute and solvent molecules are similar to each other) and dilute solutions.

- 2) **Boiling-Point Elevation:** the boiling point of the solution is higher than that of the pure solvent.

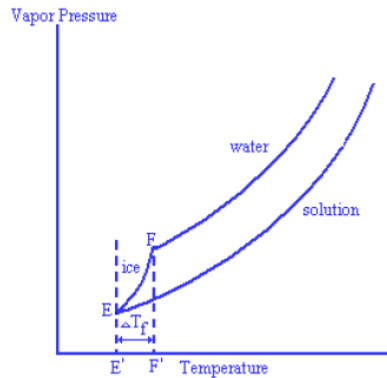


$$\Delta T_b = T_b (\text{solution}) - T_b (\text{solvent}) = i \cdot K_b \cdot m;$$

- $T_b (\text{solution})$: the boiling point of the solution;
- $T_b (\text{solvent})$: the boiling point of the pure solvent;
- m : the molality of the solute;
- K_b : the molal boiling-point-elevation constant for the solvent ($\text{K} \cdot \text{kg} \cdot \text{mol}^{-1}$);
- i : van't Hoff factor (for a nonelectrolyte, $i=1$; for an electrolyte, i will depend on how the substance ionizes in that solvent)

$\rightarrow \frac{\text{moles of particles in solution}}{\text{moles of formula units dissolved}}$

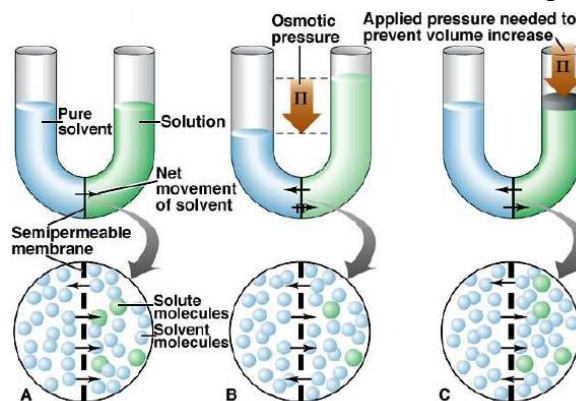
- 3) **Freezing-Point Depression:** the freezing-point of the solution is lower than that of the pure liquid.



$$\Delta T_f = T_f(\text{solution}) - T_f(\text{solvent}) = -i \cdot K_f \cdot m;$$

K_f : the molal freezing-point-depression constant ($\text{K} \cdot \text{kg} \cdot \text{mol}^{-1}$)

4) **Osmotic Pressure:** the minimum pressure which needs to be applied to a solution to prevent the inward flow of water across a semipermeable membrane.



$$\Pi = i \left(\frac{n}{V} \right) RT = iMRT;$$

- a) Π : the osmotic pressure;
- b) V : the volume of solution;
- c) i : van't Hoff factor;
- d) n : the number of moles of solute;
- e) R : ideal-gas constant;
- f) T : the absolute temperature
- g) M : the molarity of the solute
- **Colloids:** the intermediate types of dispersions are called colloids. *dissolve → separate*
 - 1) Particle size can be used to classify a mixture as colloid or solution. Colloid particles range in diameter from 5 to 1000 nm; solute particles are smaller than 5 nm in diameter.
- **Hydrophilic:** a molecule or other molecular entity that is attracted to water molecules and tends to be dissolved by water.
- **Hydrophobic:** a molecule or other molecular entity that is not attracted to water and may seem to be repelled by it. *adsorb (attach) ions → disperse in water (stable)*
- **Tyndall Effect:** the scattering of light by colloidal particles.
- **Brownian motion:** collisions that cause a random motion of the entire colloidal particle.